

Problem 01

Suppose the following a random sample of size 59 comes from an exponential distribution with mean θ . Find the minimum χ^2 estimate of θ by using R programming.

x	frequency
[0, 1)	4
[1, 2)	10
[2, 3)	24
[3, 4)	12
[4, 5)	6
[5, ∞)	3



$$Pr(a \leq x < b) = F(b) - F(a) = (1 - e^{-\frac{b}{\theta}}) - (1 - e^{-\frac{a}{\theta}}) = e^{-\frac{a}{\theta}} - e^{-\frac{b}{\theta}}$$

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

```
ni <- c(4, 10, 24, 12, 6, 3)
n <- sum(ni)

chisquare <- function(a) {
  sapply(a, function(single_a) {
    lower <- c(0, 1, 2, 3, 4, 5)
    upper <- c(1, 2, 3, 4, 5, Inf)
    p <- exp(-lower/single_a) - exp(-upper/single_a)

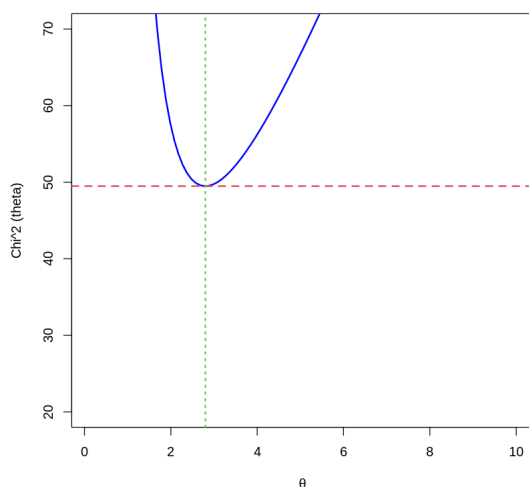
    sum((ni - n * p)^2 / (n * p))
  })
}

est <- optimize(chisquare, c(0, 10))

# Print results
cat("Estimated theta:", est$minimum, "\n")
cat("Minimum chi-square value:", est$objective, "\n")

# Plotting the chi-square function
curve(chisquare, from = 0.1, to = 10, col = "blue", lwd = 2,
      xlab = expression(theta), ylab = "Chi^2 (theta)",
      ylim = c(20, 70))
abline(h = est$objective, v = est$minimum, lwd=2, lty = c(2,3), col = c(2,3))
```

Estimated theta: 2.799088
Minimum chi-square value: 49.48542



Problem 02

Suppose the following a random sample of size 59 comes from an exponential distribution with mean $\frac{1}{\theta}$. Find the minimum χ^2 estimate of θ by using R programming.

x	frequency
[0, 1)	4
[1, 2)	10
[2, 3)	24
[3, 4)	12
[4, 5)	6
[5, ∞)	3

```
ni <- c(4, 10, 24, 12, 6, 3)
n <- sum(ni)

chisquare <- function(a) {
  sapply(a, function(single_a) {
    lower <- c(0, 1, 2, 3, 4, 5)
    upper <- c(1, 2, 3, 4, 5, Inf)
    p <- exp(-lower*single_a) - exp(-upper*single_a)

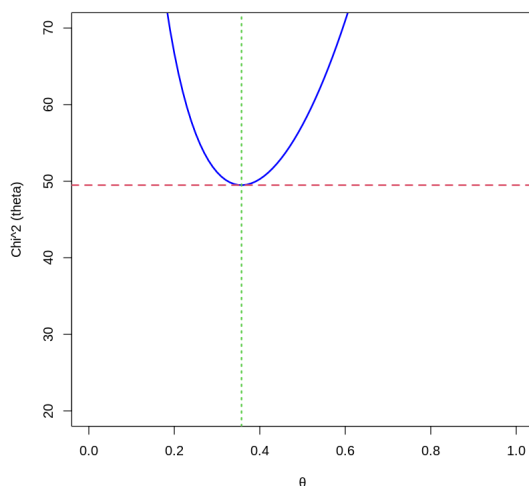
    sum((ni - n * p)^2 / (n * p))
  })
}

est <- optimize(chisquare, c(0, 10))

# Print results
cat("Estimated theta:", est$minimum, "\n")
cat("Minimum chi-square value:", est$objective, "\n")

# Plotting the chi-square function
curve(chisquare, from = 0, to = 1, col = "blue", lwd = 2,
      xlab = expression(theta), ylab = "Chi^2 (theta)",
      ylim = c(20, 70))
abline(h = est$objective, v = est$minimum, lwd=2, lty = c(2,3), col = c(2,3))
```

Estimated theta: 0.35727
Minimum chi-square value: 49.48542



Problem 03

Let the following random sample be drawn from Weibull distribution with the shape parameter $\beta = 1.5$, scale parameter α [The value α is taken from 63.2% quantile for the given random sample]

105, 140, 20, 113, 121, 10, 44, 150, 60, 30, 30, 11

- Find the MLE's of α and β with their variances.
- Construct 95% CI for α and β .
- Estimate median, 63.2th and 75th percentile.
- Comment on your findings.

$$\ell(\alpha, \beta) = n \ln\left(\frac{\beta}{\alpha}\right) + (\beta - 1) \sum_{i=1}^n \ln\left(\frac{x_i}{\alpha}\right) - \sum_{i=1}^n \left(\frac{x_i}{\alpha}\right)^{\beta}.$$

```
library(stats4)
x = c(105, 140, 20, 113, 121, 10, 44, 150, 60, 30, 30, 11)
n = length(x)

logL = function(a, b) {
  # if (a <= 0 || b <= 0) return(Inf)

  term1 = n*log(b/a)
  term2 = (b-1)*sum(log(x/a))
  term3 = - sum((x/a)^b)

  -(term1 + term2 + term3)
}

fit = mle(logL, start = list(a=quantile(x,.632),b=1.5))

coefs = coef(fit)

var = vcov(fit)

cat("MLE estimates: ", coefs, "\n")

cat("Variance are", diag(var), "\n")

### b
cat("95% confidence intervals for a (scale parameter):\n")
cat("Lower:", coefs[1] - qnorm(0.975) * sqrt(var[1, 1]), "\n")
cat("Upper:", coefs[1] + qnorm(0.975) * sqrt(var[1, 1]), "\n")

cat("95% confidence intervals for b (shape parameter):\n")
cat("Lower:", coefs[2] - qnorm(0.975) * sqrt(var[2, 2]), "\n")
cat("Upper:", coefs[2] + qnorm(0.975) * sqrt(var[2, 2]), "\n")

percentile = function(p) {
  qweibull(p, scale = coefs[1], shape = coefs[2])
}

### c
cat("Estimated median:", percentile(0.5), "\n")
cat("Estimated 63.2th percentile:", percentile(0.632), "\n")
cat("Estimated 75th percentile:", percentile(.75), "\n")
```

```
MLE estimates: 75.74329 1.324994
Variance are 303.4913 0.09845565
95% confidence intervals for a (scale parameter):
Lower: 41.59875
Upper: 109.8878
95% confidence intervals for b (shape parameter):
Lower: 0.710034
Upper: 1.939984
Estimated median: 57.43966
Estimated 63.2th percentile: 75.72456
Estimated 75th percentile: 96.91821
```

```
summary(fit)
```

```
Maximum likelihood estimation

Call:
mle(minuslogl = logL, start = list(a = quantile(x, 0.632), b = 1.5))

Coefficients:
      Estimate Std. Error
a.63.2% 75.743287 17.4210014
b      1.324994  0.3137764

-2 log L: 124.5367
```

✓ Problem 04

Let the following random sample be drawn from Gamma distribution with the shape parameter $\beta = 1.5$, scale parameter α [The value α is taken from 63.2% quantile for the given random sample]

105, 140, 20, 113, 121, 10, 44, 150, 60, 30, 30, 11

- Find the MLE's of α and β with their variances.
- Construct 95% CI for α and β .
- Estimate medium, 63.2th and 75th percentile.
- Comment on your findings.

$$\ell(\alpha, \beta) = (\alpha - 1) \sum_{i=1}^n \ln x_i - \frac{\sum_{i=1}^n x_i}{\beta} - n\alpha \ln \beta - n \ln \Gamma(\alpha).$$

```
library(stats4)

x <- c(105, 140, 20, 113, 121, 10, 44, 150, 60, 30, 30, 11)
# x <- rgamma(100, shape = 2, scale = 50) # Example data
mean(x)
n <- length(x)
logL <- function(a, b) {
  if (a <= 0 || b <= 0) return(Inf) # ensure valid values

  term1 <- n * lgamma(a)           # n * log Gamma(a)
  term2 <- n * a * log(b)          # n * a * log(b)
  term3 <- - (a - 1) * sum(log(x)) # - (a-1) * sum(log(x))
  term4 <- sum(x)/b                # sum(x) / b

  term1 + term2 + term3 + term4    # return negative log-likelihood
}

# Start values (use method of moments as a better guess)
fit <- mle(logL,
  start = list(a=quantile(x,.632),b=1.5),
  method = "L-BFGS-B",
  lower = c(0.01, 0.01)) # Ensure positive a and b

# Extract MLE and variance-covariance matrix
coefs <- coef(fit)
var <- vcov(fit)

cat("MLE estimates: ", coefs, "\n")

cat("Variance are", diag(var), "\n")

### b
cat("95% confidence intervals for a (shape parameter):\n")
cat("Lower:", coefs[1] - qnorm(0.975) * sqrt(var[1, 1]), "\n")
cat("Upper:", coefs[1] + qnorm(0.975) * sqrt(var[1, 1]), "\n")

cat("95% confidence intervals for b (scale parameter):\n")
cat("Lower:", coefs[2] - qnorm(0.975) * sqrt(var[2, 2]), "\n")
cat("Upper:", coefs[2] + qnorm(0.975) * sqrt(var[2, 2]), "\n")

percentile = function(p) {
  qgamma(p, shape = coefs[1], scale = coefs[2])
}
```

```
### c
cat("Estimated median:", percentile(0.5), "\n")
cat("Estimated 63.2th percentile:", percentile(0.632), "\n")
cat("Estimated 75th percentile:", percentile(.75), "\n")
```

```
69.5
MLE estimates: 1.512018 45.96505
Variance are 0.3161652 408.627
95% confidence intervals for a (shape parameter):
Lower: 0.4099586
Upper: 2.614078
95% confidence intervals for b (scale parameter):
Lower: 6.345306
Upper: 85.58479
Estimated median: 54.92146
Estimated 63.2th percentile: 73.19521
Estimated 75th percentile: 95.13852
```

Double-click (or enter) to edit

▼ Problem 05

Let the following random sample be drawn from **exponential distribution** with mean $\frac{1}{\theta}$. Find the 95% CI for θ using **pivotal quantity** and **Bayesian method**. Also compare between them. [$\alpha_1 = 0.025, \alpha_2 = 0.025$]

86, 146, 251, 623, 98, 175, 176, 76, 264, 15, 157, 220, 42, 321, 180, 198, 38, 20, 61, 121, 282, 224, 189, 180, 325

$$\frac{\chi_{2n}^2}{2s} \sim \text{Gamma}(\alpha = \frac{2n}{2}, \theta = \frac{1}{s})$$

$$\therefore \frac{\chi_{2n}^2}{2s} \sim \text{Gamma}(\alpha = n, \theta = \frac{1}{s})$$

$$\text{Also, } \frac{\chi_{2n}^2}{2s} \sim \text{Gamma}(\alpha = n, \beta = s)$$

θ is the scale parameter and β is the rate parameter.

▼ Theory

Here, $n = 25$ (sample size) and $s = \sum x = 4468$

Pivotal 95% CI:

For an exponential with rate θ (mean $\frac{1}{\theta}$), the sum $S = \sum_{i=1}^n X_i \sim \text{Gamma}(\text{shape} = n, \text{rate} = \theta)$

```
a1 = 0.025
a2 = 0.025

x = c(86, 146, 251, 623, 98, 175, 176, 76, 264, 15, 157, 220, 42, 321,
      180, 198, 38, 20, 61, 121, 282, 224, 189, 180, 325)

s = sum(x)
n = length(x)
```

```
# Degrees of freedom
df <- 2 * n

# Quantiles of chi-square
lower_quant = qchisq(a1, df)
upper_quant = qchisq(1-a2, df)

# Pivotal CI
theta_lower <- lower_quant / (2*s)
theta_upper <- upper_quant / (2*s)

theta_lower2 <- qgamma(a1, shape = n, rate = s)
theta_upper2 <- qgamma(1-a2, shape = n, rate = s)
```

```

CI_pivotal <- c(theta_lower, theta_upper)
cat("95% CI using pivotal:", CI_pivotal, "\n")
cat("Range of CI", theta_upper - theta_lower, "\n")
theta_lower2
theta_upper2

```

```

95% CI using pivotal: 0.003621012 0.007992412
Range of CI 0.0043714
0.0036210120518866
0.00799241217407189

```

```

# Bayesian method CI
alpha0 <- 1
beta0 <- 1

# Posterior parameters
alpha_post <- alpha0 + n
beta_post <- beta0 + s

# Credible interval
theta_bayes_lower <- qgamma(a1, shape = alpha_post, rate = beta_post)
theta_bayes_upper <- qgamma(1 - a2, shape = alpha_post, rate = beta_post)

CI_bayes <- c(theta_bayes_lower, theta_bayes_upper)

```

```

cat("95% CI using Bayesian:", CI_bayes, "\n")
cat("Range of CI", theta_bayes_upper - theta_bayes_lower, "\n")

```

```

95% CI using Bayesian: 0.003800417 0.008257984
Range of CI 0.004457567

```

```

CI_pivotal - CI_bayes

```

```

-0.000179404868139432 -0.000265572094786996

```

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```

a1 = 0.025
a2 = 0.025

x = c(86, 146, 251, 623, 98, 175, 176, 76, 264, 15, 157, 220, 42, 321,
      180, 198, 38, 20, 61, 121, 282, 224, 189, 180, 325)

s = sum(x)
n = length(x)

# Pivotal CI
theta_lower <- qgamma(a1, shape = n, rate = s)
theta_upper <- qgamma(1-a2, shape = n, rate = s)

cat("95% CI using pivotal:", theta_lower, theta_upper, "\n")
cat("Range of CI", theta_upper - theta_lower, "\n")

# Bayesian method CI
alpha0 <- 1
beta0 <- 1

# Posterior parameters
alpha_post <- alpha0 + n
beta_post <- beta0 + s

# Credible interval
theta_bayes_lower <- qgamma(a1, shape = alpha_post, rate = beta_post)
theta_bayes_upper <- qgamma(1 - a2, shape = alpha_post, rate = beta_post)

cat("95% CI using Bayesian:", theta_bayes_upper, theta_bayes_lower, "\n")
cat("Range of CI", theta_bayes_upper - theta_bayes_lower, "\n")

```

```
95% CI using pivotal: 0.003621012 0.007992412
Range of CI 0.0043714
95% CI using Bayesian: 0.008257984 0.003800417
Range of CI 0.004457567
```

Problem 06

Obtain the two-step M-estimator of the location parameter of the following ψ function.

$$\Psi(x) = \begin{cases} -k, & \text{if } x < -k, \\ x, & \text{if } -k < x < k, \\ k, & \text{if } x > k. \end{cases}$$

Also compute it's variance and compare with estimate the variance of estimator $(\bar{x})$. Given a ranodm sample from $N(\mu, \sigma^2)$:

2.1, 5.2, 2.3, 1.4, 2.2, 2.3, 1.6

[Consider $k = 2$]

$$d = \frac{\text{median}(|x - \text{median}(x)|)}{0.6745}$$

$$\psi' = \begin{cases} 1, & \text{if } -k < x < k, \\ 0, & \text{otherwise.} \end{cases}$$

$$\theta_{j+1} = \theta_j + \frac{d \times \sum \psi}{\sum \psi'}$$

$$\text{var}(\theta) = \frac{d^2 \times \sum \psi^2}{(\sum \psi')^2}$$

```
x = c(2.1, 5.2, 2.3, 1.4, 2.2, 2.3, 1.6)
k = 2

theta1 = median(x)
d = median(abs(x - theta1)) / 0.6745
h = (x - theta1)/d
psi <- ifelse(h < -k, -k,
             ifelse(h < k, h, k)
)

psi.prime = ifelse(h < -k, 0, ifelse(h > k, 0, 1))

theta2 = theta1 + d*sum(psi) / sum(psi.prime)

theta1
theta2

# 2nd
h = (x - theta2)/d
psi <- ifelse(h < -k, -k,
             ifelse(h < k, h, k)
)

psi.prime = ifelse(h < -k, 0, ifelse(h > k, 0, 1))

theta3 = theta2 + d*sum(psi) / sum(psi.prime)
theta3

var = d^2 * sum(psi^2) / (sum(psi.prime))^2
cat("robust var", var, "\n")

n = length(x)
## Variance of xbar
cat("Normal variance", var(x)/n)
```

```
2.2
2.15087101556709
2.15087101556709
robust var 0.01957785
Normal variance 0.2289796
```

Another Method

```
library(MASS)
x=c(2.1,5.2,2.3,1.4,2.2,2.3,1.6)
firstattempt=huber(x,k=2)
mu=firstattempt$mu
mu
s=firstattempt$s
secondattempt=hubers(x, k = 2, mu, s, initmu = mu, tol = 1e-06)

mu2=secondattempt$mu
sd2=secondattempt$s

cat("Mu_2", mu2, "\n")

var= sd2^2
var

xvar=var(x)/length(x)
xvar
```

```
2.15087014848924
Mu_2 2.15087
0.0219810276
0.228979591836735
```